Medium ACR
MRI Scanning Guide

For use with the Scottish Medium ACR Analysis Framework

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Introduction

This document was created to help standardise image acquisition across Scotland for MRI QA involving the Medium ACR phantom. It contains information on sequence set-up and parameter values, along with positioning help to ensure compatibility with the Scottish Medium ACR Analysis Framework.

This guidance is primarily aimed at acceptance testing and QA using the head coil but can be generalised to any coil in which the Medium ACR phantom can fit. If sites wish to use the Medium ACR phantom with coils other than the head and neck coil used clinically (e.g. additional neurovascular coils, anterior array coils and spine coils) then we suggest an axial/transverse (T1 and T2) acquisition is sufficient if acquired alongside acquisitions in all 3 orientations in the head and neck coil. For this, the phantom should be positioned centred relative to the coil and where possible all padding should be removed such that the minimum distance between phantom and the coil can be achieved.

To ensure ease of use, the parameters for the initial ACR sequence are detailed in the appendices. This is to facilitate separate instructions for each MRI scanner manufacturer currently in use in Scotland. Please follow the guide, which will direct you to the appendices at the relevant time.

This document should be used in conjunction with *Large and Medium Phantom Test Guidance for the ACR MRI Accreditation Program*.

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1 Equipment

- Medium ACR Phantom
- Large and Medium ACR Phantom Test Guidance
- MR Safe Spirit Level (recommended)
- USB Flash Drive (or alternative means of transferring data)
- Phantom Cradle (optional) or padding for positioning phantom

2 Considerations

- The phantom should be placed in the MR scanner room beforehand to acclimatise.
- The scans should be acquired on the head coil that is routinely used for clinical brain imaging.
- 'Nose' and 'Chin' markers are on the phantom to help orient positioning.
- The temperature of the scan room should be recorded on the day of scanning, to ensure comparability of data. We recommend including it in the Patient ID/Patient Name field as below.
- Refer to *Large and Medium ACR Phantom Test Guidance* as needed.

3 Set-up

Use the following parameters when setting up the patient exam:

Suggested Naming Convention	Board_Site_Scanner_Date_RoomTemp *e.g. GGC_INS_SOLA_011025_19-4C*
Date of Birth	01/01/1990
Height	160cm
Weight	90kg
Sex	Other (or if 'Other' is not available, Male)
Patient Position	Head First Supine
Region	Brain

If time permits, a set of images should be acquired for each phantom orientation (i.e. images acquired in the transverse, sagittal, and coronal planes). For each orientation, acquire:

- A standard scanner localiser
- An ACR localiser
- An ACR T1 image
- (Optional) ACR T2 image
- (Optional) Clinical T1 image
- (Optional) Clinical T2 image

For daily or weekly QA, a simple Transverse T1 acquisition is thought to be sufficient but for monthly, annual or acceptance testing, it would be expected to be acquired in all 3 orientations within the head coil. A T2 acquisition would also be expected in the axial orientation for monthly, annual or acceptance testing.

Where time permits, sites may wish to test other coils which are able to accommodate the phantom volume. These might include the bore transmit body coil, T/R head coils, neurovascular coils, anterior array coils and spine coils. For these measurements only the transverse T1 acquisition is required. For surface coils (e.g. anterior array, spine) measurements should be made with the coil as close to the

phantom as possible (e.g. with all padding removed and the coils laying directly on top). For volume coils (e.g. T/R head coil) measurements should be obtained with the phantom centralised in the coil. This is not possible for the transmit body coil where measurements should be made with the phantom on the patient couch.

4 Scanning the Medium ACR Phantom

4.1 Transverse Orientation

4.1.1 Phantom Position & Localiser

Set the phantom up in standard positioning (Figure 1), using the nose and chin markers on the phantom, along with a spirit level, to ensure phantom is positioned correctly and level. The phantom can be positioned with or without a manufacturer specific cradle.



Figure 1: Transverse positioning of Medium ACR phantom in head coil.

Align the lasers to the centre of the phantom (rather than the head coil markers), let the phantom settle, then run a standard scanner localiser. If possible, turn off any automatic alignment.

4.1.2 ACR Localiser

Bring in a very basic T1 spin echo sequence (the manufacturer's protocol tree can be useful here) to plan the ACR localiser from. An overview has been included below, but **please refer to the relevant appendix for your scanner's manufacturer** (Appendix A – Siemens, Appendix B – GE, Appendix C – Phillips, Appendix D – United Imaging) to fully plan the localiser sequence.

Localiser Orientation	Sagittal
Suggested Naming Convention	ACR_TRA_sag_loc
Pulse Sequence	Spin Echo
TR (ms)	200
TE (ms)	20
# of Slices	1
Slice Thickness (mm)	10
Slice Gap (mm)	N/A
Phase Encoding Direction	A >> P
Scan Time (min:sec)	0:56 (approx.)

All later ACR sequences will be copied from this initial sequence, so please ensure to follow the relevant appendix and select the correct parameters. Before acquiring your scan, check that the slice position is through the centre of the phantom.

4.1.3 ACR T1 Scan

Check the ACR localiser - it should look very similar to Figure 2, if not, your phantom positioning is likely incorrect.

Copy and paste the ACR localiser sequence 'ACR_TRA_sag_loc' into your protocol, and change the following parameters:

	T1-weighted
Suggested Naming Convention	ACR_TRA_T1
Orientation	Transverse/Axial
TR (ms)	500
TE (ms)	20
# of Slices	11
Slice Thickness (mm)	5
Slice Gap (mm)	5 (100%)
Phase Encoding Direction	R >> L
Scan Time (min:sec)	2:16 (approx.)

Once the sequence is set up, ensure the first and last slice intersects with the 45° wedges (as shown in Figure 2 below). Line mode can be a useful tool here, as it shows each slice in the FOV - it is often depicted by the symbol on the right or hidden under 'Options' headers.



Please note: the ACR manual will assume the slices are acquired in the direction shown in Figure 2 and refers to numbered slices in the document. However, the Analysis Framework will work regardless of the direction the slices are acquired in.

After the ACR_TRA_T1 sequence has finished acquiring, check that the orientation appears correct and is similar to Figure 3, with the resolution blocks at the bottom and the circle at the top left.

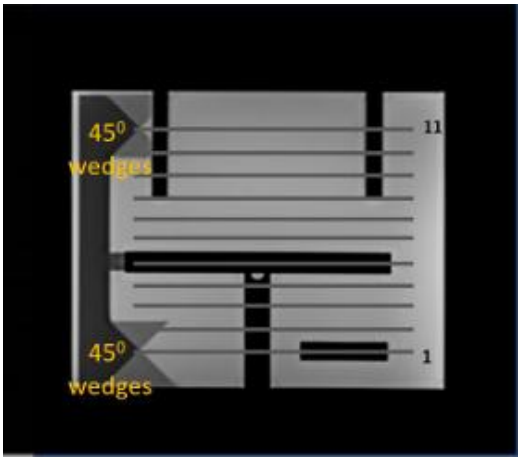


Figure 2: Image of Medium ACR phantom, acquired using an ACR localiser sequence. First and last slices in the FOV are aligned with 45° wedges. [Image taken from ACR Large and Medium Phantom Test Guidance]

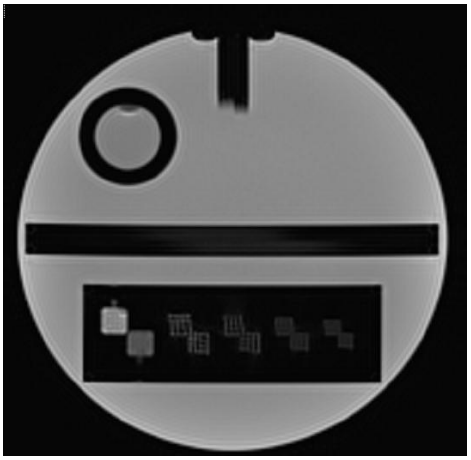


Figure 3: One slice of the Medium ACR phantom, acquired using an ACR T1 sequence in any orientation. [Image taken from ACR Large and Medium Phantom Test Guidance]

4.1.4 (Optional) ACR T2 Scan

Copy and paste the ACR_TRA_T1 sequence, adjusting the sequence parameters listed below to acquire a T2 weighted image:

	T2-weighted
Suggested Naming Convention	ACR_TRA_T2
Orientation	Transverse/Axial

TR (ms)	2000
TE (ms)	80
# of Slices	11
Slice Thickness (mm)	5
Slice Gap (mm)	5 (100%)
Phase Enc. Direction	R >> L
Scan Time (min:sec)	8:56 (approx.)

Position the slices as shown in Figure 2 and acquire the ACR_TRA_T2 sequence. After acquisition, ensure the resulting image matches Figure 3's orientation.

4.1.4.1 Phillips ACR T2-weighted scan TE/TR conflict

A common conflict was found for Phillips MRI scanners, where the required TR and TE values for the ACR T2 weighted scan could not be achieved. If you are scanning on a Phillips scanner and getting this conflict, a workaround was devised by using two echoes with the following TE values:

	First Echo	Second Echo
TE	20ms	80ms

After assigning TE values to the two echoes, turn 'Optimise second echo' to 'OFF', otherwise the bandwidths will differ for the two echoes. Once 'optimise second echo' is off, you can select the desired WFS, and this will set the same bandwidth to the two resultant image series.

Please note that the analysis script can detect when there are images present with differing TRs or TEs within a single DICOM folder and enable the user to select a specific image series with the desired TE for analysis (see Figure 4).

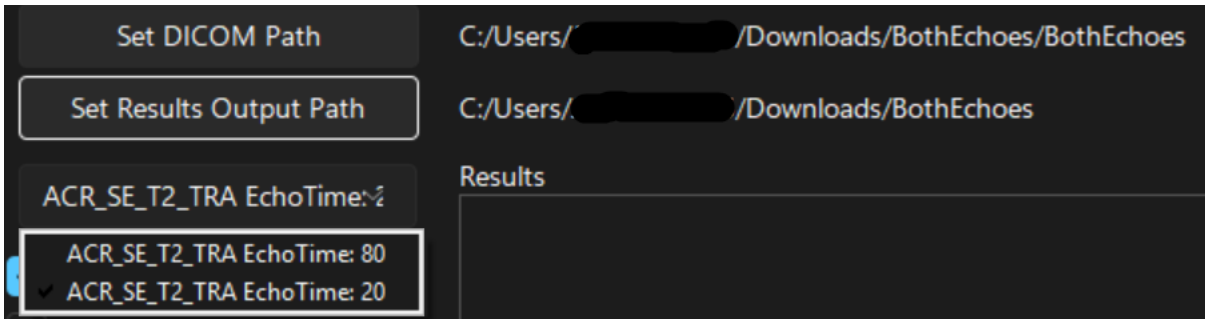


Figure 4: Screenshot of analysis script showing the multiple TEs being detected and divided so the correct TE value can be selected.

4.1.5 (Optional) Clinical T1 & T2 Scans

Bring in a site-specific clinical sequence for transverse T1-weighted and T2-weighted brain imaging to your protocol, changing only the following parameters for both acquisitions:

# of Slices	11
Slice Thickness (mm)	5
Slice Gap (mm)	5

Align the slices as shown in Figure 2, to achieve the orientation displayed in Figure 3.

Please note that both clinical scans are optional to acquire. If you are only acquiring one clinical scan, it is recommended to acquire a T1 weighted image over a T2 weighted image, due to acquisition time.

4.2 Sagittal Orientation

4.2.1 Phantom Positioning & Localiser

Position the phantom with the spirit level bar vertical on the patient's left, the nose marker pointing superiorly, and the chin marker on the patient's left (see Figure 5). Ensure the phantom position in the patient anterior-posterior direction is in the middle of the head coil.

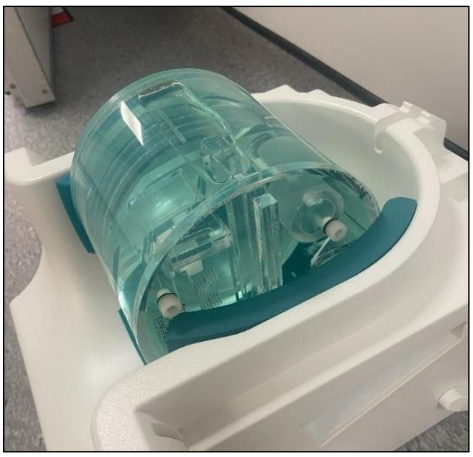


Figure 5: Sagittal positioning of Medium ACR phantom in head coil.

Align the lasers to the centre of the phantom and acquire a standard scanner localiser.

4.2.2 ACR Localiser

Copy and paste sequence 'ACR_TRA_sag_loc', changing the following parameters:

Naming Convention	ACR_SAG_cor_loc
Localiser Orientation	Coronal
Phase Enc. Direction	R >> L

Acquire the sequence, then check its appearance: the localiser should look similar to Figure 2. The image might appear rotated clockwise or anti-clockwise by 90°, but the 45° wedges should be visible. If the 45° wedges are not visible, the phantom is likely too far away from isocentre, adjusting either the phantom position or the position parameter on the scanner console.

4.2.3 ACR T1 & (Optional) ACR T2 Scans

Copy and paste the relevant sequence 'ACR_TRA_T1' or 'ACR_TRA_T2', changing the following parameters:

	T1-weighted	**T2-weighted (optional)**
Naming Convention	ACR_SAG_T1	ACR_SAG_T2
Orientation	Sagittal	Sagittal
Phase Enc. Gradient	A >> P	A >> P

Position the FOV from the recently acquired ACR localiser, using the same method as the transverse orientation scans (aligning the first and slice slices in the FOV to the 45° wedges, as illustrated in Figure 2). Acquire the scan and check the image orientation matches Figure 3.

4.3 Coronal Orientation

4.3.1 Phantom Positioning

Position the phantom with the spirit level bar going from left to right and the nose marker on the phantom facing superiorly (see Figure 6). The chin marker should be facing anteriorly. Make sure the phantom is rotated correctly in the clockwise/anti-clockwise direction. The circle component inside the phantom should be superior in the patient's anterior-superior direction, and on the patient's right.



Figure 6: Coronal positioning of Medium ACR phantom in head coil.

Align the lasers to the centre of the phantom and acquire a standard scanner localiser.

4.3.2 ACR Localiser

Copy and paste the 'ACR_TRA_sag_loc' sequence, changing only the following parameters:

Naming Convention	ACR_COR_sag_loc
Localiser Orientation	Sagittal
Phase Enc. Direction	A >> P
Slice Order	Reverse direction

Acquire the sequence and check its appearance: the localiser should look similar to Figure 2, with the 45° wedges visible. If the wedges are not visible, the phantom position will need adjusted.

4.3.3 ACR T1 & (Optional) ACR T2 Scans

Copy and paste the relevant sequence 'ACR_TRA_T1' or 'ACR_TRA_T2', changing the following parameters:

	T1-weighted	**T2-weighted (optional)**
Naming Convention	ACR_COR_T1	ACR_COR_T2
Orientation	Coronal	Coronal
Phase Enc. Gradient	R >> L	R >> L

Position the FOV using the same method as the transverse orientation scans; aligning the first and last slices with the 45° wedges as illustrated in Figure 2. Acquire the scan and check the appearance matches the orientation of Figure 3.

5 Exporting the Images

After all scans have been acquired, export to USB (do not export using Enhanced DICOM) and run the Scottish Medium ACR Analysis Framework (https://github.com/NHSH-MRI-Physics/Scottish-Medium-ACR-Analysis-Framework).

The data format is expected to be a directory containing DICOM files, where each file corresponds to one slice. It is possible to have several sequences contained within the folder when running the analysis framework.

Guidance on running the analysis is provided separately (select "Open Manual" within the analysis software).

We encourage all users of the code to upload their results to the website to collate the data obtained. To do this, select "Report Data" in the analysis software and fill out the Database Entry Form selecting the .docx file produced during the analysis.

The Data entered will be checked for conformity to this guidance and, if it complies, will be added to the following website: https://med-acr-community-data-dashboard.streamlit.app/

Appendix A – Setting up the ACR localiser: Siemens

This scanning sequence was set up on a 1.5T Sola with XA61 software by operators Blair Johnston and Eirin Beese.

This scanner protocol is for the first ACR localiser when the phantom is in the transverse (standard) orientation. First run a standard scanner localiser, so that the ACR localiser can be planned off it. Then bring in a very basic T1 spin echo sequence (the manufacturer's protocol tree can be useful here).

All later ACR sequences will be copied from this initial sequence, so please ensure to select the correct parameters. Before acquiring your scan, check that the slice position is through the centre of the phantom.

Adjust the sequence parameters as follows:

Routine:

Suggested Naming Convention	ACR_TRA_sag_loc
Pulse Sequence	Spin Echo
# of Slices	1
Distance Factor/Slice Gap (%)	N/A
Localiser Orientation	Sagittal
Phase Encoding Direction	A >> P
Phase oversampling (%)	0
FOV Read (frequency) (mm)	250
FOV Phase (mm)	250
Slice Thickness (mm)	10
TR (ms)	200
TE (ms)	20
# of Averages (NSA)	1
Concatenations	1
Auto Align	Off
Coil Elements	Head and Neck

Contrast:

MTC	Off
Magn. Preparation	None
Flip Angle(s)	90° and 180°
Fat-Water Contrast	Standard
Dark Blood	Off
Blood Suppression	Off
Contrasts	1
Reconstruction	Magnitude
Dynamic Mode	Standard
Measurements	1
Multiple Series	Off

Resolution:

Base Resolution	256
Phase Resolution (%)	100
Interpolation (if available)	Off

Artificial Intelligence (if available)	None/Off
Acceleration (if available)	Off
Phase Partial Fourier	Off
Asymmetric Echo	Allowed
Raw Filter	Off/None
Elliptical Filter	Off
Geometry/Distortion Correction	2D
Normalise / Uniformity Correction	Prescan
Image Filter	Off/None

Sequence:

RF Pulse Type	Normal/Default
Gradient Mode Type	Normal/Default
Bandwidth	150 Hz/Px

Scan time:

Scan Time (min:sec)	0:56 (approx.)

Once this has been run, return to Section 4.1.3 – ACR T1 Scan to continue the ACR Scanning Protocol.

Additional Notes

Line Mode (hidden)

This isn't relevant for the ACR localiser, but if desiring line mode to plan the ACR T1/T2 scans, it can be hidden on some Siemens scanners. There should be an 'Options' header in the top toolbar which should show line mode. If not, it might be hidden and require you to click the arrow at the top of the planning panes to show options, before "Line Mode" can be selected. Older versions of the Siemens software can also find line mode by clicking on the 'bandana man' icon to open a window with the option.

Exporting DICOM data on a Siemens scanner

When exporting the Siemens DICOM data, export using Interoperability DICOM mode rather than Siemens Enhanced DICOM mode as this is not currently compatible with the analysis script.

Appendix B – Setting up the ACR localiser: GE

This scanning sequence was set up on a 1.5T Signa Artist with software Signa_Lx1.MR30.1_R01B_2430.c by operator Jonathan Ashmore NHS Highland. Additional edits by Blair Johnston using 1.5T Artist and 3T Pioneer.

This scanner protocol is for the first ACR localiser when the phantom is in the transverse (standard) orientation. First run a standard scanner localiser, so that the ACR localiser can be planned off it. Then bring in a very basic T1 spin echo sequence (the manufacturer's protocol tree can be useful here).

All later ACR sequences will be copied from this initial sequence, so please ensure to select the correct parameters. Before acquiring your scan, check that the slice position is through the centre of the phantom.

Adjust the sequence parameters as follows:

Suggested Naming Convention	ACR_TRA_sag_loc
Pulse Sequence	2D Spin Echo
Acceleration (ARC)	Off
Acoustic Reduction	Off
Interpolation (ZIP 512)	Off

Localiser Orientation	Sagittal
FOV (mm) (frequency)	25
FOV (mm) (phase)	1
No phase wrap	1
Slice Thickness (mm)	10
Spacing (mm)	N/A
Chem SAT (Fat/Water/Blood Supression)	None
Frequency Direction	S/I
TR (ms)	200
# of Slices	1

# of TE(s) per Scan	1
TE (ms)	20
Flip Angle(s)	90° and 180°
Artificial Intelligence (if available)	None/Off
Intensity Correction	PURE
Calibration in Prescan	On
Intensity Filter	Off/None
Save Original	Off
3D Geometry Correction	off
Matrix (frequency)	256
Matrix (phase)	256
NEX	1
Bandwidth	~35.7 kHz

Shim	Auto
(3T only) RF Drive Mode	Preset
RF Pulse Type	Normal/Default
Gradient Mode Type	Normal/Default
Coil Elements	Head and Neck

Advanced tab:

CV16 Spatial Sat Type	0 – Light
CV29 SAR optimisation	0 – OFF
CV30 Apodization level	0 – Weak
CV41 PURE compensation	0

Scan time:

Scan Time (min:sec)	0:56 (approx.)

Once this has been run, return to Section 4.1.3 – ACR T1 Scan to continue the ACR Scanning Protocol.

Appendix C – Setting up the ACR localiser: Phillips

This scanning sequence was set up on a Philips Ambition S with software version 5.7.1.3 by operator Jonathan Ashmore.

This scanner protocol is for the first ACR localiser when the phantom is in the transverse (standard) orientation. First run a standard scanner localiser, so that the ACR localiser can be planned off it. Then bring in a very basic T1 spin echo sequence (the manufacturer's protocol tree can be useful here).

All later ACR sequences will be copied from this initial sequence, so please ensure to select the correct parameters. Before acquiring your scan, check that the slice position is through the centre of the phantom.

Adjust the sequence parameters as follows:

Suggested Naming Convention	ACR_TRA_sag_loc
Localiser Orientation	Sagittal
Scan Mode **Technique**	MS (Multi-Slice) SE (Spin Echo)
TR (ms)	200
TE (ms)	20
# of Slices	1
Slice Thickness (mm)	10
Slice Gap (mm)	N/A
Fold Over Direction	A >> P
Scan Time (min:sec)	0:56 (approx.)

Coil Elements	Head and Neck
Plan Align	Off
FOV (mm) (frequency × phase)	250 × 250
# of Averages (NSA)	1
Stacks	1
Packages	1
Fold-Over Suppression	NO
ACQ & Recon Voxel Size (mm) (frequency × phase)	0.98 × 0.98
Reconstruction Matrix	256*
Flip Angle(s)	90° and 180°
Gradient Mode	Default
Half Scan	Off
Partial Echo	Off
Fat/Water/Blood Suppression	Off
Geometry/Distortion Correction (removed in newer software versions)	2D
Interpolation (if available)	Off*
Acceleration (if available)	Off
Artificial Intelligence (if available)	None/Off
Uniformity	CLEAR

Bandwidth	Change ‘water-fat shift’ to ‘user-defined’, then input: • 1.5T: 1.5 pixels • 3T: 3 pixels This should yield a displayed bandwidth of 145Hz/Px
Image Filter	Weak
Modified SE	No
Acquisition Mode	Cartesian
PNS Mode	Low (default)
Postprocessing Uniformity Correction	No
Postprocessing Geometry Correction	Default
Shim **RF Shims (at 3T)**	Auto Adaptive

** In some cases, achieving a 256x256 matrix does not appear to be possible, instead we recommend an acquisition matrix of 254x256 with a reconstruction matrix of 256x256. This yields acquired and reconstructed voxel sizes of 0.98/0.98/5 mm. Equal ACQ and Recon voxel sizes corresponds to interpolation turned off.*

Once this has been run, return to Section 4.1.3 – ACR T1 Scan to continue the ACR Scanning Protocol.

Appendix D – Setting up the ACR localiser: United Imaging

This scanning sequence was set up on a UIH 680 with R001 software by operators Blair Johnston and Eirin Beese.

This scanner protocol is for the first ACR localiser when the phantom is in the transverse (standard) orientation. First run a standard scanner localiser, so that the ACR localiser can be planned off it. Then bring in a very basic T1 spin echo sequence (the manufacturer's protocol tree can be useful here).

All later ACR sequences will be copied from this initial sequence, so please ensure to select the correct parameters. Before acquiring your scan, check that the slice position is through the centre of the phantom.

Adjust the sequence parameters as follows:

Suggested Naming Convention	ACR_TRA_sag_loc
Localiser Orientation	Sagittal
Pulse Sequence	Spin Echo

Orientation	Sagittal
Position	Isocentre
Type	Parallel
Slices	1
Slice Thickness (mm)	10
Gap	0
Readout FOV	250
Phase FOV	250
Readout Resolution	256
Readout Resolution	100
Phase Encoding (PE) Direction	A > P
Phase Oversampling (OS)	0
TR (ms)	200
TE (ms)	20
Excitation Flip Angle (FA)	90°
Refocusing Flip Angle (FA)	180°
Average	1
Sections	1
Coils	Head and Neck

Table Mode	Fixed
Bandwidth (Hz/Px)	150
Average Mode	Outer
Scan Times	1
Partial Echo	None
Partial PE	None

Bandwidth (Hz/Px)	150
Flow Comp.	None
Quiet Factor	1

Gradient Mode	Standard
RF Mode	Standard
Scan Order	Sequential
Scan Direction	Default

Shimming Mode	Default
Recovery Mode	None
Fat Suppression	None

Recon Matrix	Acq. Matrix
Uniform Correction	Advanced
Distortion Correction	2D
Image Filter	Medium
K-space Filter	Normal
Filter Strength	None
Artificial Intelligence (if available)	None

Easy Scan	Off (unticked)

Scan time:

Scan Time (min:sec)	0:53 (approx.)

Once this has been run, return to Section 4.1.3 – ACR T1 Scan to continue the ACR Scanning Protocol.